

Knowing everyone's truthfulness, one's own mechanicalness and one's subordinates' codes

A proposed affirmative answer to Question 3-1-2 of S. A. Alexander, *The Theory of Several Knowing Machines* (2013)

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Status.

- **This is a proposed resolution.** It is a long hand proof, not machine-checked and without independent human review.
- **Checks so far:** one AI system wrote the construction; a second re-derived every step and wrote out the two lemmas the draft only sketched (§2, Lemmas A and B).
- **For a reader,** the key new points are the transparent stratification (§2), the contextual-collapse schema (§3, §6) and the induction G(i) (§5, §7).
- **Priority:** targeted searches found no earlier resolution. That is not a full priority check.

Source. S. A. Alexander, *The Theory of Several Knowing Machines*, doctoral dissertation, Ohio State University, 2013 (OhioLINK ETD osu1366362999):

- Definition 1.1.1;
- the ordinal theorem, Theorem 2.1.3, which is due to T. J. Carlson, *Knowledge, machines, and the consistency of Reinhardt's strong mechanistic thesis*, Ann. Pure Appl. Logic 105 (2000) 51-82;
- the stratification method of §§2.2-2.5.

The question. Let R be an r.e. well-founded relation on $\mathbb{N}$, where $j R i$ means j is subordinate to i . Question 3-1-2 asks whether it is consistent with self-doubting Epistemic Arithmetic that every agent i :

- knows the truthfulness of everyone, $K_i \text{ ucl}(K_j \varphi \rightarrow \varphi)$ for all j and φ ;
- knows codes for its subordinates, $\exists e K_i \forall x (K_j \varphi \leftrightarrow x \in W_e)$ for $j R i$; and
- knows its own mechanicalness and that of its subordinates, $K_i \exists e \forall x (K_j \varphi \leftrightarrow x \in W_e)$ for $j = i$ and for $j R i$.

The dissertation proves only the variant (3-1-2)' (its Theorem 2.5.11). There, knowledge of a non-subordinate's truthfulness is restricted to formulas in which K_i is not exposed. Note 16 of this repository records a gap in the printed proof of that theorem. The theorem below implies it.

Answer (proposed). **Yes**, for every r.e. well-founded R , with a uniformly r.e. family. In it each machine also knows its own truthfulness. Knowledge of the subordinates' mechanicalness follows from knowledge of their codes.

1. Background

We use the base logic, the notation φ^s , consequence $\models$ and intended structures exactly as in note 14 §1.

Ordinals. The following is Theorem 2.1.3 of the dissertation, from Carlson (2000).

- For ordinals, $\alpha \leq_1 \beta$ iff $\alpha \leq \beta$ and, for all finite $X \subseteq \alpha$ and finite $Y \subseteq [\alpha, \beta)$, there is a finite $\tilde{Y}$ with $X < \tilde{Y} < \alpha$ and $X \cup \tilde{Y} \cong X \cup Y$ as structures for $(\leq, \leq_1)$.
- There is a recursive ordinal $\kappa > 0$ such that $\leq$ and $\leq_1$ are recursive on $\kappa \cdot \omega$, and $\kappa \cdot m \leq_1 \kappa \cdot n$ whenever $0 < m \leq n < \omega$.

2. Transparent i-stratification

Fix an agent i .

The language. L_i extends L_ω by operators K_i^α for $\alpha < \kappa \cdot \omega$; the unstratified K_i stays in the language.

Stratified formulas.

- A formula is **i-stratified** if two conditions hold. First, it contains no unstratified K_i . Second, whenever K_i^β occurs anywhere inside the argument of K_i^α , **including inside the argument of another operator K_k ($k \neq i$)**, we have $\beta < \alpha$.
- The other operators are unstratified but *transparent*: their arguments may contain indexed K_i . The dissertation's i-stratified formulas do not allow indices under other operators.
- A formula is **very i-stratified** if every index is a positive multiple of κ .
- **Erase** θ^- replaces every K_i^α by K_i , at every depth.

Shapes and stratifiers.

- The *shape* of a formula is obtained by replacing every variable occurrence and every term by a placeholder. Alphabetic variants, and numeral or variable substitution instances, share a shape.
- A **(partial) shape assignment** b is a function from a set S of shapes of formulas $K_i\psi$ to ordinals below $\kappa \cdot \omega$. It must satisfy two conditions:
 - S is closed under K_i -subshapes: the shapes of occurrences $K_i\chi$ lying anywhere inside ψ , including under other operators;
 - $b(\sigma) > b(\sigma')$ for every proper K_i -subshape σ' of σ .
- If b is defined on the K_i -shapes of an original formula φ , then φ^b replaces each occurrence $K_i\psi$ by $K_i^{b(\text{shape}(K_i\psi))}\psi^b$, recursively, also under other operators.
- A *stratifier* is a total computable shape assignment.
- The **height stratifier** h sets $h(\sigma) = \kappa \cdot (1 + d(\sigma))$, where $d(\sigma)$ is the nesting depth of K_i in σ . It is very.

Properties (immediate from the definitions):

- (i) φ^b is i-stratified;
- (ii) $(\varphi^b)^- = \varphi$;
- (iii) $\varphi \mapsto \varphi^b$ commutes with renaming bound variables, with substituting variables for variables (substitutability is preserved), and with numeral instances: $(\varphi^s)^b = (\varphi^b)^s$;
- (iv) if β is an order-isomorphism between finite sets of ordinals, then $\beta \circ b$ is again an assignment.

Lemma A (extension). Let b be a finite assignment with values below a limit ordinal $\lambda \leq \kappa \cdot \omega$. Then b extends to a stratifier with all values below λ .

Proof. Enumerate all shapes of K_i -formulas in order of size. For each new shape σ , set $b(\sigma) = 1 + \max\{b(\sigma') : \sigma' \text{ a proper } K_i\text{-subshape of } \sigma\}$, which is defined, or 0 if σ has none.

- Subshapes are smaller, so they have already been assigned.
- The value stays below λ because λ is a limit.

- Computability holds because the ordinal notation is recursive. $\square$

Lemma B (validity transfer).

- (1) If Φ is a valid L_ω -formula and b is a stratifier, then Φ^b is valid.
- (2) If Ψ is a valid L_i -formula containing no unstratified K_i , then Ψ^- is valid.

Proof.

- (1) Let N be any L_i -structure. Define an L_ω -structure N'' with the same first-order part: $N'' \models K_i \psi[s]$ iff $N \models (K_i \psi)^b[s]$, and $N'' \models K_k \psi[s]$ iff $N \models K_k \psi^b[s]$ for $k \neq i$.
 - N'' is a structure. Free variables are preserved by b . Alphabetic variants go to alphabetic variants, and substitution of variables commutes with b , by (iii). So the three conditions of Definition 1.2.3 transfer from N .
 - Induction on formulas gives $N'' \models \chi[s]$ iff $N \models \chi^b[s]$.
 - Since Φ is valid, $N'' \models \Phi$, hence $N \models \Phi^b$.
- (2) Given an L_ω -structure N , define N' by $N' \models K_i^\alpha \psi[s]$ iff $N \models K_i \psi^-[s]$, and $N' \models K_k \psi[s]$ iff $N \models K_k \psi^-[s]$. Erasure commutes with renaming and substitution, so N' is a structure, and $N' \models \chi[s]$ iff $N \models \chi^-[s]$. $\square$

3. The theories

A **one-hole context** $C[\cdot]$ is a formula with one distinguished subformula position, possibly under quantifiers, connectives, indexed K_i , or other operators K_k .

For each agent i and parameter n , let $\Sigma_i(n)$ consist of the universal closures of the following i -stratified formulas.

1. **Strativalidity:** $K_i^\alpha \theta$ for valid θ .
2. **Stratideduction:** $K_i^\alpha (\theta \rightarrow \psi) \rightarrow K_i^\alpha \theta \rightarrow K_i^\alpha \psi$.
3. **Stratrospection:** $K_i^\alpha \theta \rightarrow K_i^\beta K_i^\alpha \theta$ for $\alpha < \beta$.
4. **PA:** the PA axioms, with induction for i -stratified formulas.
5. **Assigned Validity:** θ^s for valid θ .
6. **Own stage truth:** $K_i^\alpha \theta \rightarrow \theta$.
7. **Other-agent truth:** $K_k \theta \rightarrow \theta$ for $k \neq i$. Here θ is any i -stratified formula; θ may expose K_i^α .
8. **Own stage mechanicalness:** $\exists e \forall x (K_i^\alpha \theta \leftrightarrow x \in W_e)$ for θ with $FV(\theta) = \{x\}$.
9. **Contextual collapse (CC):** $C[K_i^\alpha \theta] \leftrightarrow C[K_i^\beta \theta]$ for $\alpha \leq \beta$.
10. **Subordinate codes:** for $j \neq i$, every original φ with $FV(\varphi) = \{x\}$, and every finite assignment b on the K_i -shapes of φ ,

$$\forall x (K_j \varphi^b \leftrightarrow x \in W_{c(n,j,\varphi)}), \quad W_{c(n,j,\varphi)} = \{m : \langle \ulcorner \varphi \urcorner, j, m \rangle \in W_n\}.$$

The index depends only on the **original** φ , not on b .

11. **Prefix closure:** $K_i^\alpha \sigma$ for every σ generated by items 1-11 whose indices are all below α .

Let $T_i(n) = \Sigma_i(n)^-$. Both families are uniformly r.e., since R is r.e. and $\kappa \cdot \omega$ is recursive.

The fixed point. By the recursion theorem, choose q so that W_q is the set of triples $\langle \ulcorner \varphi \urcorner, j, m \rangle$ with $FV(\varphi) = \{x\}$ and $T_j(q) \models \varphi(x|\vec{m})$. Write Σ_i, T_i for $\Sigma_i(q), T_i(q)$. Then

$$m \in W_{c(q,j,\varphi)} \iff T_j \models \varphi(x|\vec{m}). \quad (1)$$

Let $\Sigma_i \cap \alpha$ be the set of axioms of Σ_i all of whose indices are below α .

The erased theory contains everything asked for.

- **Truthfulness:** T_i contains unrestricted reflection for every agent. Items 6 and 7 erase to $K_j \varphi \rightarrow \varphi$ for **every** original φ , using the lift φ^h .
- **Codes:** T_i contains the particular codes of item 10.
- **Own mechanicalness:** item 8, applied to $\theta = \varphi^h$, erases to self-mechanicalness.
- **The EA axioms and their known versions:** these come from items 1-5 and 11. Lemma B(1) gives a valid lift of every original validity, and item 11 supplies the K_i -prefixes.

4. Syntactic lemmas (no soundness assumed)

Uniformity. If β is a $(\leq, \leq_1)$ -isomorphism between finite sets of ordinals, applying β to the indices of a generator gives a generator. Each schema needs checking:

- the side conditions $\alpha < \beta$ and $\alpha \leq_1 \beta$ are preserved;
- a renamed valid formula is valid, by renaming the interpretations;
- in item 10, $\beta \cdot b$ is an assignment (property iv) and the index is unchanged;
- item 11 follows by induction.

(5) Finite-support collapse. If $\alpha \leq_1 \beta$ and θ is i -stratified with all indices below α , then $\Sigma_i \cap \alpha \models \theta$ iff $\Sigma_i \cap \beta \models \theta$.

- *Proof.* Take finitely many premises from $\Sigma_i \cap \beta$. Let X be their indices below α together with those of θ , and let Y be the premises' indices in $[\alpha, \beta)$. Choose $\tilde{Y}$ as in the definition of $\leq_1$, and apply the isomorphism $X \cup Y \rightarrow X \cup \tilde{Y}$, which fixes X .
- The renamed premises are generators (uniformity) with indices below α . The implication stays valid, and θ is unchanged. $\square$

Very counterparts. Mapping the finite set of indices of a generator order-preservingly onto positive multiples of κ gives a very generator with the same erasure. All the side conditions are "positive" ($<, \leq_1$), and they hold among κ -multiples.

(6) Equivalence of very lifts. If ρ and σ are very i -stratified and $\rho^- = \sigma^-$, then $\Sigma_i \models \text{ucl}(\rho \leftrightarrow \sigma)$.

- *Proof.* ρ and σ have the same formula tree. Fix target labels $\kappa \cdot N_p$ on the positions p of K_i -occurrences. Every N_p exceeds all indices of ρ and σ , and N_p decreases strictly along nesting.
- Change ρ 's labels one occurrence at a time, outermost first.
 - Every intermediate formula is i -stratified: ancestors already carry larger targets, and descendants still carry smaller old labels.
 - Each step is a CC instance, since $\kappa \cdot a \leq_1 \kappa \cdot N_p$.
- Do the same for σ , and chain the biconditionals. $\square$

This holds inside other operators because CC allows holes there. It does **not** need Σ_i to contain another agent's deduction axioms.

(7) Proof stratification. For very i -stratified sentences θ , $\Sigma_i \models \theta$ iff $T_i \models \theta^-$.

- **($\Rightarrow$)** Erase a valid implication from premises in Σ_i (Lemma B(2)).
- **($\Leftarrow$)**
 - Start from a valid $\sigma_1^- \rightarrow \dots \rightarrow \theta^-$ with each $\sigma_l \in \Sigma_i$.
 - Its height lift is valid (Lemma B(1)), with very components $(\sigma_l^-)^h$ and $(\theta^-)^h$.
 - Replace each $(\sigma_l^-)^h$ by the very counterpart of σ_l , and $(\theta^-)^h$ by θ , using (6). $\square$

(8) Prefix. If $\Sigma_i \cap \alpha \models \theta$ for a sentence θ , $K_i^\alpha \theta$ is i -stratified, and $\beta > \alpha$, then $\Sigma_i \cap \beta \models K_i^\alpha \theta$. This uses item 11 on the premises, item 1 on the implication, and item 2.

5. The provisional structures and the induction

Let $P = (P_k)$ be predicates on closed L_ω -sentences that are invariant under alphabetic variants. Define F_P , on standard arithmetic, by recursion on formula complexity:

$$F_P \models K_k \theta[s] \iff P_k(\theta^s) \text{ and } F_P \models \theta[s]. \quad (9)$$

F_P is a structure and is numeral-instance invariant, as in note 14. Let D_i consist of i and every agent reachable from i by a finite descending R -path. We prove by well-founded induction:

G(i). For every such P with $P_k(\sigma) \iff T_k \models \sigma$ for all $k \in D_i$, we have $F_P \models T_i$.

Nothing is assumed about P_k for $k \notin D_i$.

Fix i , assume $G(j)$ for every $j R i$, and fix P as in $G(i)$. Expand F_P to L_i as M :

$$\begin{aligned} M \models K_k \theta[s] &\iff P_k((\theta^-)^s) \text{ and } M \models \theta[s] & (k \in \mathbb{N}, \text{ unstratified } K_k), \\ M \models K_i^\alpha \theta[s] &\iff \Sigma_i \cap \alpha \models \theta^s \text{ and } M \models \theta[s]. \end{aligned} \quad (10)$$

On L_ω this is exactly F_P . The erasure in the first clause matters: the raw predicate sees the same argument whatever the indices. M is a numeral-invariant structure.

6. Contextual collapse is true in M

We claim that $M \models \text{ucl}(C[K_i \wedge \alpha \theta] \leftrightarrow C[K_i \wedge \beta \theta])$ whenever $\alpha \leq_1 \beta$ and both sides are i -stratified. The proof is by induction on the context C .

- **Empty context:** the consequence conditions agree by (5), and the truth conditions are identical.
- **Connectives and quantifiers:** immediate from the induction hypothesis.
- **Context $K_k C'$ with $k \neq i$:** both arguments have the same erasure, so the P_k conditions in (10) coincide. The truth conditions agree by induction.
- **Context $K_i^\gamma C'$:** both sides are i -stratified, so α, β and all indices of C' are below γ .
 - The CC instance for C' lies in $\Sigma_i \cap \gamma$, and Assigned Validity yields its numeral instances. So $\Sigma_i \cap \gamma$ proves one argument's numeral instance iff it proves the other's.
 - The truth conditions agree by induction.

No soundness of Σ_i is used here.

7. Subordinate codes, by rebuilding the truth filters

Fix $j R i$, an original φ with $\text{FV}(\varphi) = \{x\}$, and a finite assignment b .

- **The reconstructed predicate.** Extend b to a stratifier g (Lemma A with $\lambda = \kappa \cdot \omega$). Define $Q_k = P_k$ for $k \neq i$ and $Q_i(\theta) \iff M \models (K_i \theta)^g$. By (iii), Q_i is invariant under alphabetic variants.
- **The rebuilt structure.** Let $N = F_Q$. **Every** truth filter is rebuilt; none is frozen.

Claim. For original θ , $N \models \theta[s] \iff M \models \theta^g[s]$. The proof is by induction.

- For $K_k \theta$ with $k \neq i$: the raw tests agree because $(\theta^g)^- = \theta$, and the truth tests agree by induction.
- For $K_i \theta$: $N \models K_i \theta[s]$ iff $M \models (K_i \theta)^g[s]$ and $M \models \theta^g[s]$, using (iii) and numeral invariance. The second conjunct is implied by the first, since $(K_i \theta)^g = K_i^{g(\cdot)} \theta^g$ is factive in M .

R is well-founded, so $i \notin D_j \subseteq D_i$. Q therefore agrees with the proof predicates on D_j , and G(j) gives $N \models T_j$. Hence, by the claim and (1),

$$M \models K_j \varphi^b[x \mapsto m] \iff N \models K_j \varphi[x \mapsto m] \iff T_j \models \varphi(x|\bar{m}) \iff m \in W_{c(q,j,\varphi)}.$$

So every item-10 axiom is true in M. The unchanged predicates for D_j are *proved* correct in the rebuilt structure by G(j). No lemma that freezes other operators is used; compare note 16.

8. Stage soundness

We show $M \models \Sigma_i \cap \alpha$ by induction on $\alpha < \kappa \cdot \omega$.

- **Items 1-5:** validity, modus ponens, (8) with the truth half, standard arithmetic, and numeral invariance.
- **Items 6 and 7:** the truth halves of (10).
- **Item 9:** §6.
- **Item 10:** §7.
- **Item 8 at $\beta < \alpha$:** by induction $\Sigma_i \cap \beta$ is sound, so the extension of $K_i^\beta \theta$ is $\{m : \Sigma_i \cap \beta \models \theta(\bar{m})\}$, which is r.e.
- **Item 11 at $\beta < \alpha$:** $\sigma \in \Sigma_i \cap \beta$ is proved, and it is true by induction.

Hence $M \models \Sigma_i$.

9. Erasure, and the end of the proof

Claim. For every very i-stratified θ , $M \models \theta[s] \iff M \models \theta^-[s]$. The proof is by induction.

- At $K_k \psi$, the raw tests coincide and the truth tests agree by induction.
- At $K_i^{Kn} \psi$:
 - by cofinality and (5), $\Sigma_i \cap \kappa n \models \psi^s$ iff $\Sigma_i \models \psi^s$;
 - by (7), this holds iff $T_i \models (\psi^-)^s$, which is P_i since $i \in D_i$;
 - the truth conjuncts agree by induction.

So the clause coincides with the one for $K_i \psi^-$.

This identity is syntactic apart from the inductive truth conjunct.

Conclusion of G(i). Every axiom of T_i is the erasure of a generator with a very counterpart. That counterpart is true in M, so the erased axiom is true in M, hence in F_P . So $F_P \models T_i$, which is G(i).

The theorem. Take P_k to be T_k -provability for every k. Then G(i) holds for every i, so every filter is redundant, and F_P is the intended structure of the uniformly r.e. family. §3 lists the required contents of T_i .

Theorem (proposed). For every r.e. well-founded R there is a uniformly r.e. family of knowing machines satisfying self-doubting Epistemic Arithmetic, together with:

- unrestricted mutual truthfulness;
- known particular codes of subordinates;
- known own mechanicalness.

In particular, Question 3-1-2 has the answer yes. ■

10. Remarks

- **Why this does not settle 3-1-3.** Question 3-1-3 also asks for knowledge of the mechanicalness of **non-subordinates**. G(i) is proved for *arbitrary* external predicates, and under such predicates an external mechanicalness axiom can be false. For example, if P_k accepts every sentence, its filtered column on $x \notin W_x$ is the non-r.e. nonhalting set. 3-1-3 remains open.
- **The two ingredients that make exposed reflection possible.**
 - *Transparency*: indices may occur under other operators, and other operators test the erased argument. This makes item 7 free of charge even when K_i is exposed.
 - *Contextual collapse*: CC erases to tautologies, so it gives no machine another machine's private code axioms, and it is proved true directly in §6.